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TALEN ENERGY

MAY 0 5 2017

U. S. Nuclear Regulatory Commission
Attn: Document Control Desk
Washington, DC 20555-0001

10 CFR 50.73

SUSQUEHANNA STEAM ELECTRIC STATION
LICENSEE EVENT REPORT 50-387(388)/2017-003-00
UNIT 1 LICENSE NO. NPF-14 **Docket No. 50-387**
UNIT 2 LICENSE NO. NPF-22 **50-388**
PLA-7604

Attached is Licensee Event Report (LER) 50-387(388)/2017-003-00. This LER reports an event involving the loss of Secondary Containment due to a low flow trip of all Zone III fans. This LER is being submitted pursuant to the requirements of 10 CFR 50.73(a)(2)(v)(C) as a condition that could have prevented fulfillment of a safety function.

Should you have any questions regarding this submittal, please contact Mr. Jason Jennings, Manager – Nuclear Regulatory Affairs at (570) 542-3155.

There were no actual consequences to the health and safety of the public as a result of this event.

This letter contains no new regulatory commitments.

B. Berryman

Attachment: LER 50-387(388)/2017-003-00

Copy: NRC Region I
Ms. T. E. Hood, NRC Project Manager
Ms. L. Micewski, NRC Sr. Resident Inspector
Mr. M. Shields, PA DEP/BRP

LICENSEE EVENT REPORT (LER)

(See Page 2 for required number of digits/characters for each block)

(See NUREG-1022, R.3 for instruction and guidance for completing this form http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/)

1. FACILITY NAME	2. DOCKET NUMBER	3. PAGE
Susquehanna Steam Electric Station Unit 1	05000387	1 OF 3

4. TITLE Loss of secondary Containment Zone 3 Due to Fan Trip

5. EVENT DATE			6. LER NUMBER			7. REPORT DATE			8. OTHER FACILITIES INVOLVED	
MONTH	DAY	YEAR	YEAR	SEQUENTIAL NUMBER	REV NO.	MONTH	DAY	YEAR	FACILITY NAME: Susquehanna Steam Electric Station Unit 2	DOCKET NUMBER: 050000388
03	08	2017	2017	003	00	05	05	2017	FACILITY NAME	DOCKET NUMBER: 05000

9. OPERATING MODE	11. THIS REPORT IS SUBMITTED PURSUANT TO THE REQUIREMENTS OF 10 CFR §: *(Check all that apply)*			
1	☐ 20.2201(b)	☐ 20.2203(a)(3)(i)	☐ 50.73(a)(2)(ii)(A)	☐ 50.73(a)(2)(viii)(A)
	☐ 20.2201(d)	☐ 20.2203(a)(3)(ii)	☐ 50.73(a)(2)(ii)(B)	☐ 50.73(a)(2)(viii)(B)
	☐ 20.2203(a)(1)	☐ 20.2203(a)(4)	☐ 50.73(a)(2)(iii)	☐ 50.73(a)(2)(ix)(A)
	☐ 20.2203(a)(2)(i)	☐ 50.36(c)(1)(i)(A)	☐ 50.73(a)(2)(iv)(A)	☐ 50.73(a)(2)(x)
10. POWER LEVEL	☐ 20.2203(a)(2)(ii)	☐ 50.36(c)(1)(ii)(A)	☐ 50.73(a)(2)(v)(A)	☐ 73.71(a)(4)
	☐ 20.2203(a)(2)(iii)	☐ 50.36(c)(2)	☐ 50.73(a)(2)(v)(B)	☐ 73.71(a)(5)
100	☐ 20.2203(a)(2)(iv)	☐ 50.46(a)(3)(ii)	☒ 50.73(a)(2)(v)(C)	☐ 73.77(a)(1)
	☐ 20.2203(a)(2)(v)	☐ 50.73(a)(2)(i)(A)	☐ 50.73(a)(2)(v)(D)	☐ 73.77(a)(2)(i)
	☐ 20.2203(a)(2)(vi)	☐ 50.73(a)(2)(i)(B)	☐ 50.73(a)(2)(vii)	☐ 73.77(a)(2)(ii)
		☐ 50.73(a)(2)(i)(C)	☐ OTHER	Specify in Abstract below or in NRC Form 366A

12. LICENSEE CONTACT FOR THIS LER

LICENSEE CONTACT	TELEPHONE NUMBER *(Include Area Code)*
Nicole Pagliaro, Licensing Specialist – Nuclear Regulatory Affairs	570-542-6578

13. COMPLETE ONE LINE FOR EACH COMPONENT FAILURE DESCRIBED IN THIS REPORT

CAUSE	SYSTEM	COMPONENT	MANU-FACTURER	REPORTABLE TO EPIX	CAUSE	SYSTEM	COMPONENT	MANU-FACTURER	REPORTABLE TO EPIX

14. SUPPLEMENTAL REPORT EXPECTED		15. EXPECTED SUBMISSION DATE	MONTH	DAY	YEAR
☐ YES *(If yes, complete 15. EXPECTED SUBMISSION DATE)*	☒ NO				

ABSTRACT *(Limit to 1400 spaces, i.e., approximately 15 single-spaced typewritten lines)*

On March 8, 2017 at approximately 0239 hours, Secondary Containment Zone III (Unit 1&2 Reactor Building) differential pressure lowered to less than the Technical Specification (TS) limit of 0.25 in w.g. due to due to Load Center Breaker 2B240-021 (supply breaker to 2Y246) being returned to service without Motor Control Center (MCC) Breaker 2B246-023 (supply breaker to 2Y246) restored. As a result, TS 3.6.4.1 Surveillance Requirement (SR) 3.6.4.1.1 was not met due to the loss of vacuum.

The condition is being reported in accordance with 10 CFR 50.73(a)(2)(v)(C) as an event or condition that could have prevented fulfillment of a safety function.

The cause of the event was less than adequate procedure use and adherence.

Corrective actions included a communication to Operators and a revision to the Procedure and Work Instructions Use and Adherence procedure to perform a second check for non-conditional procedure steps.

There were no actual consequences to the health and safety of the public as a result of this event.

1. FACILITY NAME	2. DOCKET NUMBER	3. LER NUMBER		
		YEAR	SEQUENTIAL NUMBER	REV NO.
Susquehanna Steam Electric Station Unit 2	05000-387	2017	- 003	- 00

NARRATIVE

CONDITIONS PRIOR TO EVENT

Unit 1 – Mode 1, approximately 100 percent Rated Thermal Power
Unit 2 – Mode 5, approximately 0 percent Rated Thermal Power

There were no structures, systems, or components that were inoperable at the start of the event that contributed to the event. Unit 2 was in a refueling outage and Zone II was relaxed.

EVENT DESCRIPTION

On March 8 at approximately 0239, during work to replace Load Center Breaker 2B240-021 (supply breaker to 2B246) [EIIS: BKR] and inspect MCC Breaker 2B246-023 [EIIS: BKR], both units Zone III HVAC [EIIS:NA] differential pressure was lost when Operations commenced the restoration phase prematurely. The error occurred when Load Center Breaker 2B240-021 (supply breaker to 2B246) [EIIS: BKR] was returned to service without MCC Breaker 2B246-023 [EIIS: BKR] restored. Unit 2 Zone III isolation dampers [EIIS:DMP] were temporarily aligned to an alternate power supply during the work. When Operations attempted to restore Zone 3 HVAC damper control power to normal, there was no power available from Distribution Panel 2Y246. The Unit 2 division II Zone III isolation dampers [EIIS: DMP] failed to their safety "closed" position causing a low flow trip of all Unit 2 Zone III fans [EIIS:FAN].

This resulted in Zone III differential pressure lowering to less than the TS 3.6.4.1 allowable limit of 0.25 in w.g.

A timeline of events is as follows:

0100: MCC 2B246 de-energized by opening Load Center Breaker 2B240-021 [EIIS: BKR] which is the supply breaker to MCC 2B246 [EIIS: BKR].

0145: Load Center Breaker 2B240-021 [EIIS: BKR] was closed per work release RLWO 1873012 to energize MCC 2B246 [EIIS: BKR] and utilize the RB elevator.

0200: Replacement breaker arrived at work site. Load Center Breaker 2B240-021 [EIIS: BKR] was opened per work release RLWO 1873012 to de-energize MCC 2B246 [EIIS: BKR] and continue job.

0230: Operations team directed that MCC 2B246 could be re-energized while work continued on MCC Breaker 2B246-023 [EIIS: BKR].

0235: Commenced MCC 2B246 bus restoration.

0239: Operations place the Control Power Supply to Unit 2 RB Dampers to normal. Unit 2 Zone III Division II dampers [EIIS: DMP] closed due to loss of control power causing Unit 2 Zone III fans [EIIS:FAN] to trip on low flow.

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Susquehanna Steam Electric Station Unit 2	05000-387	2017	- 003	- 00

CAUSE OF EVENT

The cause of the event was less than adequate procedure use and adherence.

ANALYSIS/SAFETY SIGNIFICANCE

Secondary Containment Zone III pressure was not maintained above the TS 3.6.4.1 required vacuum of 0.25 in w.g. with respect to outside atmosphere. The event was caused by a loss of power to the Unit 2 Zone III isolation dampers, which caused them to close. This resulted in the shutdown of the Unit 2 Reactor Building fans and a loss of Zone III vacuum. This event is being reported pursuant to 10 CFR 50.73(a)(2)(v)(C) as a condition that could have prevented fulfillment of a safety function to mitigate the consequences of an accident by controlling the release of radioactive material.

There was no actual safety consequence as a result of this event. Engineering analysis of this event has determined that secondary containment could have performed its safety function of isolating, as assumed in the accident analysis, and also of re-establishing 0.25 in w.g. vacuum (drawdown) within the assumed accident analysis time (10 minutes). This event will not be counted as a safety system functional failure (SSFF) for the NRC performance indicator.

CORRECTIVE ACTIONS

Corrective actions included a communication to Operators and a revision to the Procedure and Work Instructions Use and Adherence procedure to perform a second check for non-conditional procedure steps.

PREVIOUS SIMILAR EVENTS

LER 50-387(88)/2016-003-00, "Unit 2 Zone 3 HVAC unable to maintain Zone 3 dP >0.25 in wg," dated May 6, 2016 (PLA-7459)

LER 50-387(388)/2015-002-00, "Loss of Secondary Containment Due to Failure of Running Fans," dated June 10, 2015 (PLA-7329)